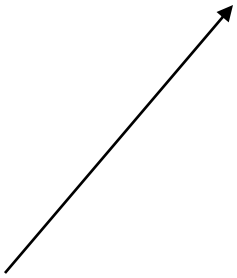
Compiled Problems – Senior Division

1. **11-12 C1 Assembly Language**

What is the final value of N when the program is run?



```
A     DC      32
B     DC      2
N     DC      0
TOP   LOAD    A
      DIV     B
      STORE   C
      LOAD    N
      ADD     =1
      STORE   N
      LOAD    C
      STORE   A
      BG      TOP
      LOAD    N
      SUB     =1
      STORE   N
      PRINT   N
      END
```

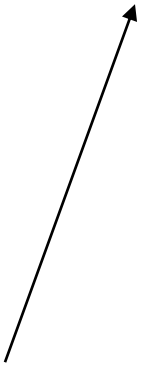
2. **12-13 C1 Assembly Language**

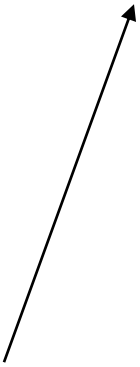
What is the final value of S that is printed when the program is run?

```
S     DC      0
N     DC      10
TOP   LOAD    N
      MULT    N
      ADD     S
      STORE   S
      LOAD    N
      SUB     =1
      STORE   N
      BG      TOP
      PRINT   S
      END
```

3. **13-14 C1 Assembly Language**

What is the final value of E when the program is run?





```
A     DC      10
B     DC      2
C     DC      5
D     DC      8
      LOAD    A
      SUB     B
      BG      P
R     LOAD    B
      MULT    C
      SUB     D
      BL      S
T     LOAD    A
      SUB     D
      BE      U
V     LOAD    A
      DIV     B
      ADD     C
      MULT    D
      STORE   E
      LOAD    E
      PRINT   E
      END
P     LOAD    A
      SUB     B
      STORE   A
      BU      R
S     LOAD    B
      ADD     C
      STORE   B
      BU      T
U     LOAD    D
      SUB     D
      STORE   D
      BU      V
```

4. 14-15 C1 Assembly Language

What is printed when the program is run?

```
X       DC      21      LOAD    K               BU      DONE
Y       DC      20      ADD     L       RSLT    LOAD    T
Z       DC      23      STORE   M               PRINT   T
T       DC      1       LOAD    Z       DONE    END
F       DC      0       MULT    Z
        LOAD    X       STORE   N
        MULT    X       LOAD    M
        STORE   K       SUB     N
        LOAD    Y       BE      RSLT
        MULT    Y       LOAD    F
        STORE   L       PRINT   F
```

5. 15-16 C1 Assembly Language

What is printed when the program is run?

```
A       DC      2       LOAD    D
B       DC      1       ADD     = 1
C       DC      1       STORE   D
D       DC      1       LOAD    C
TOP     LOAD    B       SUB     = 100
        MULT    A       BL      TOP
        STORE   B       PRINT   D
        LOAD    C       END
        ADD     B
        STORE   C
```